AI-USE DISCLOSURE

Quantum Traffic Gridlock | Separate disclosure

Purpose

The challenge statement permits AI assistants and general-purpose libraries, while requiring assistance to be declared and results to be independently verified.

1. AI tools used

Claude was used for programming assistance during development of the project. ChatGPT was used to help interpret the challenge requirements, troubleshoot Python dependency and file-path errors, review the generated output artifacts, and draft the technical report and this disclosure.

2. Scope of assistance

The team should make this description more specific if possible: identify which parts of the event-specific code or formulation were suggested, generated, edited, or debugged with Claude, and whether any other AI tools were used. Do not claim that either AI tool independently verified the full implementation unless that verification was actually performed.

3. Human/team responsibility

The team remains responsible for understanding, checking, and submitting the traffic model, QUBO construction, QAOA circuit, optimization procedure, and reported results. The supplied implementation includes a traffic simulator, second-order QUBO fitting, QAOA optimization, an OpenQASM 3 exporter, and output generation. This statement describes the supplied files; it does not independently establish who authored each component.

4. Verification status and limitations

The report's numerical claims are limited to the supplied problem statement and reference-scenario R output files. Hidden/shared scenario H results, five-seed robustness, independent Qiskit execution, and complete 256-schedule surrogate-regret validation were not supplied and remain to be completed. Before submission, independently verify the official scorer checks, QUBO prediction error and regret over all 256 schedules, OpenQASM parsing and bit ordering, 4,096-shot count consistency, reproducibility, runtime, and results for R and H.